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Configure Jetty to Log HTTP Requests

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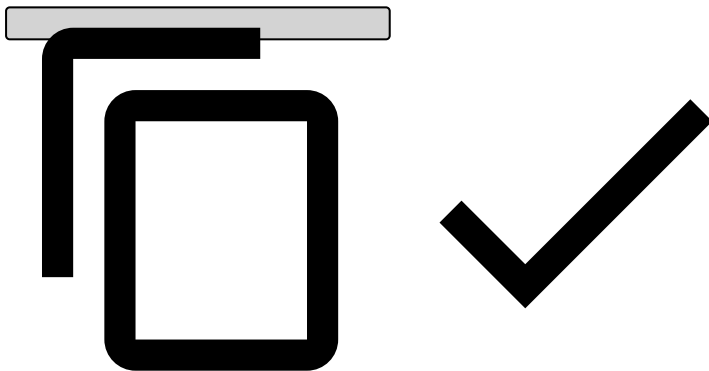
Pulse uses Jetty **9.4.X** as the back-end engine for the web server. It is possible to configure the engine to automatically log every HTTP request into a log file (which will appear in the `log/` folder). To do so, it's necessary to add a new logger entry to the file `conf/engine/jetty/jetty.xml` in Pulse:

```
<Configure id="Server" class="org.eclipse.jetty.server.Server">
  <Arg name="threadpool"><New id="threadpool" class="org.eclipse.jetty.util.thread.QueuedThreadPool"/></Arg>
  ...
  <Set name="RequestLog">
    <New id="RequestLog" class="org.eclipse.jetty.server.CustomRequestLog">
      <!-- Writer -->
```

```

<Arg>
  <New class="org.eclipse.jetty.server.AsyncRequestLogWriter">
    <!-- Make http request log rotate by day. -->
    <Arg><Property name="solr.log.dir" default="log"/>/>yyyy_mm_dd.request.log</Arg>
    <Set name="filenameDateFormat">yyyy_MM_dd</Set>
    <!-- Only delete rotated files after a week. -->
    <Set name="retainDays">
      <Property name="solr.log.requestlog.retaindays" default="7"/>
    </Set>
    <Set name="append">true</Set>
    <Set name="timeZone">UTC</Set>
  </New>
</Arg>
<!-- Default format String: -->
<!-- Arg><Get class="org.eclipse.jetty.server.CustomRequestLog"
name="EXTENDED_NCSA_FORMAT"/></Arg -->
<!-- Custom format String: https://javadoc.io/doc/org.eclipse.jetty/jetty-server/
9.4.51.v20230217/org/eclipse/jetty/server/CustomRequestLog.html -->
<Arg>{{date: %{YYYY-MM-dd:hh:mm:ss:SSSZ}t},{response status:%s},{hostname: %{client}a},
{response time: %{ms}Tms}, {hostname: %{Host}i}, {user: %u}, {Connection status: %X}, {response
status: %s}, {first line: %r}, {address client: %{client}a}, {address server: %{server}a}, {MAC:
%{local}a}}</Arg>
  </New>
</Set>
...

```



The API of CustomRequestLog for **Jetty v9.4.51.v20230217** (used by **Pulse 25.0.15+**) is described at [the jetty-server documentation](https://javadoc.io/doc/org.eclipse.jetty/jetty-server/9.4.51.v20230217/org/eclipse/jetty/server/CustomRequestLog.html).

By using this CustomRequestLog it's possible to satisfy the following requirements from [CR707 PCR4293 Fraud Event Logging Components_Requirements](#):

Requirement	Loggable	RequestLog Argument (string format)
Properly formatted and accurate timestamp	(tick) Must be in a format supported by Jetty DateCache.	{date: %{YYYY-MM-dd:hh:mm:ss:SSSZ}t}
Status code for the event type request was successful or not	(tick) It's possible to know via the HTTP response code.	{response status:%s}
Device identifier (MAC address or other unique identifier)	(tick) Likely not possible since printing physical address is the same as printing the IP of the client or server in our tests.	{MAC: %{local}a} It prints the IP as physical address.
Session / Transaction ID		

	(tick) Likely not possible since API doesn't provide a way to get an unique identifier for the request.	
Autonomous System Number (hostname)	(tick) Possible. Could be necessary to print the host header for it.	<pre>{hostname: %{Host}i}</pre> Might be possible for client to send wrong information.
Source IP (IPv4)	(tick)	<pre>{address client: %{client}a}</pre>
Source IP (IPv6)	(tick) Likely possible since IPv4 works, but untested.	<pre>{address client: %{client}a}</pre> It should also print IPv6 if used. Not tested.
Status Code	(tick)	<pre>{response status: %s}</pre>
Response Time	(tick)	<pre>{response time: %{ms}Tms}</pre>
Additional headers (i.e., HTTP headers)	(tick)	<pre>{Header_Name: %{Header_Name}i}</pre>
Where appropriate, the username and/or userID shall be included	(tick) Likely not possible since API doesn't provide a way to get an unique identifier for the request.	
Where appropriate, the command executed shall be included	(tick)	<pre>{first HTTP line: %r}</pre>
Where possible, all data shall be formatted as key-value-pairs allowing for easy extraction	(tick)	E.g.: <pre>{{date: %{YYYY-MM-dd:hh:mm:ss:SSSZ}t},{response status:%s},{hostname: %{client}a}, {response time: %{ms}Tms}, {hostname: %{Host}i}, {user: %u}, {Connection status: %X}, {response status: %s}, {first line: %r}, {address client: %{client}a}, {address server: %{server}a}, {MAC: %{local}a}}</pre>
Where possible, a unique event identifier shall be included for event correlation; a unique event identifier shall be defined per event type6	(tick) Likely not possible since API doesn't provide a way to get an unique identifier for the request.	

The third column of the table above shows examples of possible values for the argument `formatString` that needs to be passed to the constructor of `org.eclipse.jetty.server.CustomRequestLog#CustomRequestLog( org.eclipse.jetty.server.RequestLog.Writer, java.lang.String )` in order to print the requirements of the first column.